Now we prove II. We can compute that

$$\nabla^2_{\theta',\theta}\mathcal{H}_2(\theta,\theta') = \nabla_{\theta'}\nabla_\theta \mathbb{E}_{x\sim\pi_\gamma p_\theta}[\log p_{\theta'}(x)] \tag{25}$$

$$= \nabla_{\theta'}\nabla_\theta \int_{x\in\mathbb{R}^d} \log p_{\theta'}(x)\left(\frac{p_\theta(x)+\gamma p_{\theta^\star}(x)}{1+\gamma}\right)dx \tag{26}$$

$$= \frac{1}{1+\gamma}\nabla_{\theta'}\nabla_\theta \int_{x\in\mathbb{R}^d} \log p_{\theta'}(x)p_\theta(x)dx \tag{27}$$

$$= \frac{1}{1+\gamma}\nabla^2_{\theta',\theta}\mathbb{E}_{x\sim p_\theta}[\log p_{\theta'}(x)] \tag{28}$$

$$= \frac{1}{1+\gamma}B \tag{29}$$

where the third equality holds because the integral containing $p_{\theta^\star}$ is constant with respect to θ. Next, we can compute that

$$B = \int_X \nabla_{\theta'}\log p_{\theta'}(x)\nabla_\theta p_\theta(x)dx\Big|_{\theta^*,\theta^*} \tag{30}$$

$$= \int_X [\nabla_\theta \log p_\theta(x)][\nabla_\theta p_\theta(x)]dx\Big|_{\theta^*,\theta^*} \tag{31}$$

$$= \int_X \nabla_\theta[p_\theta(x)\nabla_\theta\log p_\theta(x)]dx\Big|_{\theta^*,\theta^*} - \int_X p_\theta(x)(\nabla_\theta\nabla_\theta\log p_\theta(x))dx\Big|_{\theta^*,\theta^*} \tag{32}$$

$$= \int_X \nabla_\theta\left[p_\theta(x)\frac{\nabla_\theta p_\theta(x)}{p_\theta(x)}\right]dx\Big|_{\theta^*,\theta^*} - \nabla^2_{\theta',\theta'}\mathbb{E}_{x\sim p_\theta}[\log p_{\theta'}(x)]\Big|_{\theta^*,\theta^*} \tag{33}$$

$$= -C, \tag{34}$$

where the third equality follows from the product rule for gradients,

$$\nabla_\theta[p_\theta(x)\nabla_\theta\log p_\theta(x)] = p_\theta(x)(\nabla_\theta\nabla_\theta\log p_\theta(x)) + [\nabla_\theta p_\theta(x)][\nabla_\theta\log p_\theta(x)]. \tag{35}$$

Finally, we will prove the formula (23) by manipulating (22). We begin with the rightmost factor in (22). If we apply these equalities that we just obtained, then we get

$$\begin{aligned}
\mathcal{J}(\pi\mathcal{G}(\theta^\star)) &= -\left(\nabla^2_{\theta',\theta'}\mathcal{H}(\theta^\star,\theta^\star)\right)^{-1}\cdot\lambda\nabla^2_{\theta',\theta}\mathcal{H}_2(\theta^\star,\theta^\star)\\
&= -(A+\lambda C)^{-1}\cdot\frac{\lambda}{1+\gamma}B\\
&= -(I+\lambda A^{-1}C)^{-1}\cdot\frac{\lambda}{1+\gamma}A^{-1}B\\
&= (I+\lambda A^{-1}C)^{-1}\cdot\frac{\lambda}{1+\gamma}A^{-1}C
\end{aligned}$$

where the first equality follows from (23) along with the fixed point Proposition A.6, and we are using that A is invertible by Assumption 2 A.4, which implies all eigenvalues of A are nonzero; in the fourth step we used that $B = -C$. This proves part II.

Now we prove III. We can bound the operator norm $\|A^{-1}C\|$ as follows:

$$\|A^{-1}C\| = \|I + A^{-1}(C-A)\| \le \|I\| + \|A^{-1}\|\cdot\|C-A\| \le 1 + \alpha^{-1}\|C-A\|, \tag{36}$$

where the first estimate comes from subadditivity and submultiplicativity, and the second comes from the fact that, since A is symmetric, $\|A\| = \max_{\lambda\in\sigma(A)}|\lambda|$, where $\sigma(A)$ is the spectrum of A. Formally, we know by Assumption A.4 that A has eigenvalues $e_1 < e_2 < \cdots < e_n \le -\alpha < 0$ and so $|e_n| > \alpha$. Therefore, A^{-1} has eigenvalues $1/e_n < 1/e_{n-1} < \cdots < 1/e_1 < 0$ and thus $1/|e_n| > 1/|e_{n-1}| > \cdots > 1/|e_1|$, which gives us the bound $\|A^{-1}\| = 1/|e_n| < 1/\alpha$ on the matrix

norm. Next, we can estimate that

$$\begin{aligned}||C - A|| &= \|\nabla^2_{\theta',\theta'}\mathbb{E}_{x\sim p_{\theta^\star}}[\log p_{\theta'}(x)]|_{\theta^\star} - \nabla^2_{\theta',\theta'}\mathbb{E}_{x\sim p_{\text{data}}}[\log p_{\theta'}(x)]|_{\theta^\star}\| \\ &= \|\mathbb{E}_{x\sim p_{\theta^\star}}[\nabla^2_{\theta',\theta'}\log p_{\theta^\star}(x)] - \mathbb{E}_{x\sim p_{\text{data}}}[\nabla^2_{\theta',\theta'}\log p_{\theta^\star}(x)]\| \\ &\leq L d_W(p_{\theta^\star}, p_{\text{data}}) \\ &= L\varepsilon,\end{aligned}$$

where in the second equality we exchange the derivative and the expectation in equation 4 using the Dominated Convergence Theorem, since Assumption 1 A.3 says that $x \mapsto \nabla_\theta \nabla_\theta \log p_\theta(x)$ is L-Lipschitz; and in the last estimate, we used Kantorovich-Rubenstein duality. This, combined with the estimate (36), yields the bound in (24). □

We are finally ready to prove our theorem that guarantees convergence to the optimal parameters in the infinite sampling case under certain assumptions, one being the that the initial model parameters θ_0 are sufficiently close to $\theta^\star$:

Theorem A.8 (Convergence of Iterative Fine-tuning, Infinite Sampling Case)**.** *Suppose we have an iterative fine-tuning procedure defined by the rule $\theta^\infty_{t+1} = \pi_\gamma \mathcal{G}^\infty_\lambda(\theta^\infty_t)$. Let $\theta^\star$ be the parameter vector for the optimal generative model, as in (9). We assume that $\theta^\star$ follows Assumptions A.3 and A.4 from (Bertrand et al., 2024). Suppose also that $\lambda\left(1 + \frac{\varepsilon L}{\alpha}\right) < \frac{1+\gamma}{2+\gamma}$. Then, the Jacobian of $\pi_\gamma G^\infty_\lambda$ satisfies the following bound:*

$$\|\nabla_\theta \pi_\gamma \mathcal{G}^\infty_\lambda(\theta^\star)\|_2 \leq \frac{1}{1+\gamma} \cdot \frac{\lambda(\alpha + \varepsilon L)}{\alpha - \lambda(\alpha + \varepsilon L)} < 1. \tag{37}$$

Consequently, there exists a $\delta > 0$ such if $\theta_0 \in \Theta$ satisfies $\|\theta_0 - \theta^\star\| \leq \delta$, then starting training at θ_0 and having $\theta_{t+1} = \pi_\gamma \mathcal{G}^\infty_\lambda(\theta_t)$, we have that $\lim_{t\to\infty} \theta_t \to \theta^\star$. Furthermore, if we define

$$\rho(\lambda) = \frac{\lambda(\alpha + \varepsilon L)}{\alpha - \lambda(\alpha + \varepsilon L)}, \tag{38}$$

then we obtain the asymptotic stability estimate[3]

$$\|\theta_t - \theta^\star\| \leq \left(\frac{\rho(\lambda)}{1+\gamma}\right)^t \|\theta_0 - \theta^\star\|. \tag{39}$$

Proof. **We first prove the Jacobian bound** (37)**.** By hypothesis, we know $\lambda(1 + \frac{L\varepsilon}{\alpha}) < 1$, so by Lemma A.7(III), we have $\lambda||A^{-1}C|| < 1$. Thus, we can write

$$(I + \lambda A^{-1}C)^{-1} = \sum_{k=0}^{\infty}(-\lambda A^{-1}C)^k$$

and so

$$\|(I + \lambda A^{-1}C)^{-1}\| \leq \sum_{k=0}^{\infty} \lambda^k ||A^{-1}C||^k = \frac{1}{1 - \lambda||A^{-1}C||}.$$

Applying Lemma A.7(2), we get

$$||\mathcal{J}(G(\theta^\star))|| \leq ||(I + \lambda A^{-1}C)^{-1}|| \cdot \frac{\lambda}{1+\gamma}||A^{-1}C|| \leq \frac{\lambda}{1+\gamma} \cdot \frac{||A^{-1}C||}{1 - \lambda||A^{-1}C||}.$$

Now, it is straightforward to see the RHS above is at most the bound in (37) if and only if $\alpha\|A^{-1}C\| < \alpha + \varepsilon L$. But this bound holds because of Lemma A.7(III). This proves the Jacobian bound (37), but does not prove that the bound is less than 1. For this, we must show that

$$\frac{1}{1+\gamma} \cdot \frac{\lambda(\alpha + \varepsilon L)}{\alpha - \lambda(\alpha + \varepsilon L)} < 1. \tag{40}$$

[3](Bertrand et al., 2024) could have presented their results in this stronger form, without the big O notation, with very little extra work.

By clearing denominators and grouping like terms, we can see that this is equivalent to

$$\lambda\left(1+\frac{\varepsilon L}{\alpha}\right)<\frac{1+\gamma}{2+\gamma}, \tag{41}$$

which is precisely guaranteed by our hypothesis.

We now apply the the Jacobian bound (37) to prove the asymptotic stability estimate (39). Assume λ is sufficiently small so that $\rho(\lambda)/(1+\gamma)<1$. Then for every $\rho'\in(\rho(\lambda)/(1+\gamma),1)$, there exists $\delta>0$ sufficiently small so that every $\theta_0\in\Theta$ which satisfies $\|\theta_0-\theta^\star\|<\delta$ has the property that $\|\nabla_\theta\pi_\gamma G_\lambda^\infty(\theta_0)\|_2<\rho'$. Because the map $\pi_\gamma G_\lambda^\infty$ has Jacobian matrix norm less than 1 in the δ-ball around $\theta^\star$, it is a contraction mapping in this neighborhood. Concretely, this means that

$$\|\pi_\gamma\mathcal{G}_\lambda^\infty(\theta)-\pi_\gamma\mathcal{G}_\lambda^\infty(\theta')\|\leq\frac{\rho(\lambda)}{1+\gamma}\|\theta-\theta'\|, \tag{42}$$

for every θ,θ' in the δ-ball around $\theta^\star$. In particular, for $(\theta,\theta')=(\theta_t,\theta^\star)$ we obtain

$$\|\theta_{t+1}-\theta^\star\|=\|\pi_\gamma\theta_t-\theta^\star\|=\|\pi_\gamma\mathcal{G}_\lambda^\infty(\theta_t)-\pi_\gamma\mathcal{G}_\lambda^\infty(\theta^\star)\|\leq\frac{\rho(\lambda)}{1+\gamma}\cdot\|\theta_t-\theta^\star\|.$$

By induction, the above estimate implies that if θ_0 is in a δ-ball around $\theta^\star$, then so is every successive θ_t. Therefore the desired estimate (39) now follows by induction on t. □

Remark A.9. Taking $\gamma=0$ recovers exactly the result in (Bertrand et al., 2024). Importantly, the correction function π_γ provides leverage in determining how large the augmentation percentage λ can be: choosing a larger correction strength γ allows us to choose a larger augmentation percentage λ while still retaining theoretical guarantees for convergence. Additionally, for the same choice of augmentation percentage λ, a larger correction strength γ provides a guarantee for an improved rate of convergence. See Conjecture 4.7.

A.3. Stability of Iterative Fine-tuning with Correction for Finite Sampling

Finally, we prove a stability result for iterative fine-tuning with correction in the presence of statistical error. To do this, we require an assumption that essentially provides probabilistic guarantee that the chosen generative model learns the underlying distribution increasingly better if it has access to more samples:

Assumption A.10. There exist $a,b,\varepsilon_{\text{OPT}}\geq 0$ and a neighborhood U of $\theta^\star$ such that, for any $\delta\in(0,1)$, with probability $1-\delta$ over the samplings, we have

$$(\forall\theta\in U)(\forall n\in\mathbb{N})\qquad\|\pi_\gamma\mathcal{G}_\lambda^n(\theta)-\pi_\gamma\mathcal{G}_\lambda^\infty(\theta)\|\leq\varepsilon_{\text{OPT}}+\frac{a}{\sqrt{n}}\sqrt{\log\frac{b}{\delta}}. \tag{43}$$

See Appendix B for a discussion about this assumption; we investigated whether to assume a similar bound to the one they assumed in (Bertrand et al., 2024), or prove our bound from theirs. In fact, we prove in Appendix B that you can in fact deduce something nearly as strong as Assumption A.10 from Assumption 3 in their paper, so we made Assumption A.10 for the sake of a cleaner, more parallel exposition.

Theorem A.11 (Iterative Fine-Tuning Stability Under Correction)**.** *Suppose we have an iterative fine-tuning procedure defined by the rule $\theta_{t+1}^n=\pi_\gamma\mathcal{G}_\lambda^n(\theta_t^n)$. In words, this means that the augmentation percentage is $\lambda\in(0,\infty)$ and the correction strength is $\gamma\in[0,\infty)$. Under the same assumptions of Theorem A.8 and Assumption A.10, there exist $0<\rho<1$ and $\delta_1>0$ such that if $\|\theta_0^n-\theta^\star\|\leq\delta_1$, then for any $\delta_2\in(0,1)$, with probability $1-\delta_2$, we have*

$$\|\theta_t^n-\theta^\star\|\leq\left(\varepsilon_{OPT}+\frac{a}{\sqrt{n}}\sqrt{\log\frac{bt}{\delta}}\right)\sum_{i=0}^{t}\left(\frac{\rho(\lambda)}{1+\gamma}\right)^i+\left(\frac{\rho(\lambda)}{1+\gamma}\right)^t\|\theta_0^n-\theta^\star\|. \tag{44}$$

Proof. By the triangle inequality, we can estimate that

$$\begin{aligned}\|\theta_t^n-\theta^\star\|&\leq\|\theta_t^n-\pi_\gamma\mathcal{G}_\lambda^\infty(\theta_{t-1}^n))\|+\|\pi_\gamma\mathcal{G}_\lambda^\infty(\theta_{t-1}^n)-\theta^\star\|\\&=\|\pi_\gamma\mathcal{G}_\lambda^n(\theta_{t-1}^n)-\pi_\gamma\mathcal{G}_\lambda^\infty(\theta_{t-1}^n)\|+\|\pi_\gamma\mathcal{G}_\lambda^\infty(\theta_{t-1}^n)-\pi_\gamma\mathcal{G}_\lambda^\infty(\theta^\star)\|,\end{aligned} \tag{45}$$